

Focus Bear – Automated App Analysis System

Detailed Project Documentation & Instructions

## 1. Overview

The Automated App Analysis System is an end-to-end analytics platform designed to analyse productivity apps across the market using ETL pipelines, machine learning, NLP/LLM processing, and multi-page interactive dashboards. It automates data collection, cleaning, modelling, scoring, and insight generation.

## 2. Project Structure

Core Folders:

- app.py – Main Streamlit dashboard with interactive analytics.
- etl/ – Extract–Transform–Load pipelines for raw app data.
- scrapers/ – App Store & Google Play scrapers.
- llm/ – OpenAI-powered summarisation and insight engines.
- data/raw – Input JSON/CSV scraped app data.
- data/curated – Processed, analysis-ready data tables.
- writes/ – Output reports (PDF, CSV, JSON summaries).

## 3. Dashboard Features

Page 1 – App Overview:

- Displays rating, ranking, install count, sentiment trends.
- Auto-generates market insights with OpenAI.

Page 2 – Review Explorer:

- Review filtering, keyword cloud, sentiment charts.
- Export options for filtered datasets.

Page 3 – Sentiment Analytics:

- Positive/neutral/negative breakdown.
- Time trends + theme clustering.

Page 4 – Comparison Dashboard:

- Side-by-side competitor analytics for Focus Bear.
- Feature matrix across both platforms.

Page 5 – Productivity Insights:

- Behaviour patterns, user journeys, retention themes.
- AI-driven insight summaries.

Page 6 – Data Tools:

- Trigger ETL refresh, run scrapers, run AI processors.

## **4. Running the Application**

Local Setup:

### **1. Clone:**

```
git clone https://github.com/Focus-Bear-analytics/automated-app-analysis-system
```

## **2. Install deps:**

```
pip install -r requirements.txt
```

## **3. Add .env:**

```
OPENAI_API_KEY=YOUR_KEY
```

## **4. Run Dashboard:**

```
streamlit run app.py
```

## **5. Adding New App Data**

Steps:

### **1. Drop raw JSON into:**

```
data/raw/playstore/
```

```
data/raw/appstore/
```

### **2. Run ETL:**

```
python etl/run_etl.py
```

### **3. Refresh dashboard.**

## **6. Maintaining the Project**

Scrapers:

`python scrapers/play_scraper.py`

`python scrapers/apple_scraper.py`

LLM Model Updates:

Modify:

`llm/summarizer.py`

`llm/insight_engine.py`

## **7. Exporting Reports**

Dashboard can export:

- PDF insights
- Structured CSV analytics
- JSON summaries

All stored in `writes/exports/`.

## **8. Troubleshooting**

Streamlit not loading → Clear `.streamlit` cache.

API errors → Check API key.

Missing data → Re-run ETL.

## **9. Contribution Workflow**

```
git checkout -b feature/new-feature
```

```
git add .
```

```
git commit -m "your message"
```

```
git push origin feature/new-feature
```

## **10. License**

Internal proprietary analytics tool for Focus Bear.